DIABETES PREDICTION USING LOGISTIC REGRESSION

NAME

Radha Krishan Tripathi

COLLEGE

LNCT University

PROJECT NAME

Diabetes Prediction using Logistic Regression

01 | Project Overview

The objective of this project is to build a binary classification model that predicts whether a person is likely to have diabetes based on diagnostic measurements. The target variable is **Outcome**, where 0 represents no diabetes and 1 represents diabetes.

OBJECTIVE	DATASET	MODEL
Predict diabetes outcome (0 or 1).	Pima Indians Diabetes Database from Kaggle; 768 records and 8 input features.	Median imputation, StandardScaler and balanced Logistic Regression.

Dataset and Input Features

The dataset contains 768 observations and the following eight input features: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction and Age. The target column is Outcome.

Feature	Description
Pregnancies	Number of pregnancies.
Glucose	Glucose measurement.
BloodPressure	Blood pressure measurement.
SkinThickness	Skin thickness measurement.
Insulin	Insulin measurement.
BMI	Body mass index.
DiabetesPedigreeFunction	A function representing diabetes pedigree information.
Age	Age in years.

Workflow

Load the Kaggle CSV → inspect the data → identify invalid zero values → replace those values with missing values → split the data into training and testing sets → perform median imputation and feature scaling → train balanced Logistic Regression → evaluate the model → interpret the coefficients.

02 | Data Preparation and Evaluation

Data Inspection and Cleaning

The dataset contained 768 rows and 9 columns, with no pre-existing null values and no duplicate records. Some zero values in medical measurement columns were treated as invalid measurements. Zero values were retained for Pregnancies and Outcome because zero is a valid value for those columns.

Column	Zero values identified	Treatment
Glucose	5	Replaced with missing values
BloodPressure	35	Replaced with missing values
SkinThickness	227	Replaced with missing values
Insulin	374	Replaced with missing values
BMI	11	Replaced with missing values

After replacing invalid zeros with missing values, the data was split into 80% training data (614 records) and 20% testing data (154 records), using stratification to preserve the target-class proportions. Missing values were filled using the median calculated from the training data, and StandardScaler was used for feature scaling. The preprocessing steps were fitted on training data and then applied to test data.

Model Evaluation

A Logistic Regression model with balanced class weights was trained. The following metrics were obtained on the test set:

Metric	Result	Meaning
Accuracy	0.734	Proportion of all test predictions that were correct.
Precision	0.603	Reliability of positive (diabetes) predictions.
Recall	0.704	Proportion of actual positive cases detected.
F1-score	0.650	Harmonic balance of precision and recall.
ROC-AUC	0.813	Ability to distinguish between the two classes across thresholds.

Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	TN = 75	FP = 25
Actual 1	FN = 16	TP = 38

03 | Key Findings and Conclusion

Key Findings

The model achieved a ROC-AUC of 0.813 and detected approximately 70.4% of the positive cases in the fixed test split. The confusion matrix shows 75 true negatives, 25 false positives, 16 false negatives and 38 true positives. Recall is relevant in a screening-style classification task because false negatives represent positive cases that the model did not detect.

Coefficient Interpretation

The fitted model's largest positive coefficients were for Glucose and BMI. Coefficients are shown below in descending order. Because the features were standardized, the values represent changes in model log-odds per one standardized feature unit.

Feature	Coefficient
Glucose	+1.1835
BMI	+0.7096
Pregnancies	+0.3729
DiabetesPedigreeFunction	+0.2877
Age	+0.1864
SkinThickness	+0.0140
BloodPressure	-0.0146
Insulin	-0.0446

Limitation

This small historical dataset represents a specific population. Its results may not generalize to other populations or settings. The model is suitable for learning classification concepts and is not intended for medical diagnosis.

Conclusion

In this project, a balanced Logistic Regression model was developed to predict diabetes using the Pima Indians Diabetes dataset. Invalid zero measurements were handled through median imputation, and features were standardized before model training. The model achieved 73.4% accuracy, a 0.704 recall and a 0.813 ROC-AUC on the test set. Glucose and BMI had the strongest positive model coefficients. This project provided practical experience in data inspection, preprocessing, classification and model evaluation.

Educational use only: This model is not a medical diagnostic tool and should not be used to make healthcare decisions.